experts.py

`src/moe_pruning/models/olmoe/experts.py · 155 lines · module moe_pruning.models.olmoe.experts`

What it does

Expert containers for OLMoE's fused weight layout: a compact container for pruned layers, a per-expert view and a standalone reference expert.

How it works

- `PrunedFusedExperts` is a drop-in replacement for `OlmoeExperts` with the same forward signature. It stores the retained experts' rows (`gate_up_proj`, `down_proj`), their original ids (`retained_expert_ids`) and the replacement modules of removed experts (`replaced.{e}`), keyed by original router index.
- `compact()` builds the container from an original or already pruned container (so pruning twice works).
- `FusedExpertView` computes one expert from a container row without owning parameters.
- `OlmoeReferenceExpert` is a single original expert, used by the evaluation probe.

Notes

- The expert function is SwiGLU: down(act(gate) * up) with `[gate; up]` stacked in `gate_up_proj`.

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`class FusedExpertView`	Callable view of one row of a fused expert container (owns no parameters).
`.forward(x)`	Compute this expert's output from its row of the fused weights.
`class OlmoeReferenceExpert`	A single standalone original expert (used to measure reconstruction errors).
`.forward(x)`	Compute the expert's output (SwiGLU).
`class PrunedFusedExperts`	Drop-in replacement for `OlmoeExperts` with some experts removed.
`.compact(experts, remove)`	New container without the experts in `remove`.
`.retained_ids()`	Original ids of the retained experts, in row order.
`.removed_ids()`	Original ids of removed experts.
`.expert(expert_id)`	Module for an expert id: a row view if retained, the replacement if removed.
`.forward(hidden_states, top_k_index, top_k_weights)`	Run every routed (token, slot) through its expert and add gate * output.

Connections

Used by: `src/moe_pruning/models/olmoe/adapter.py`